

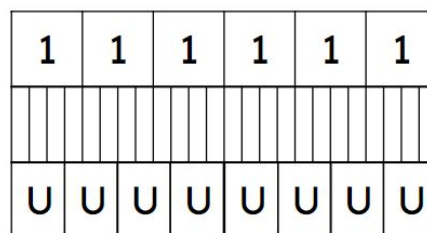
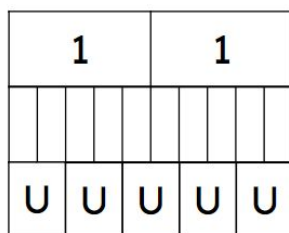
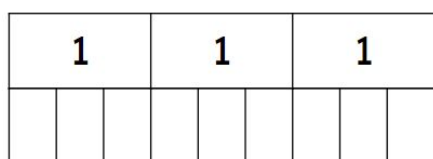


Dividing a whole number by a fraction

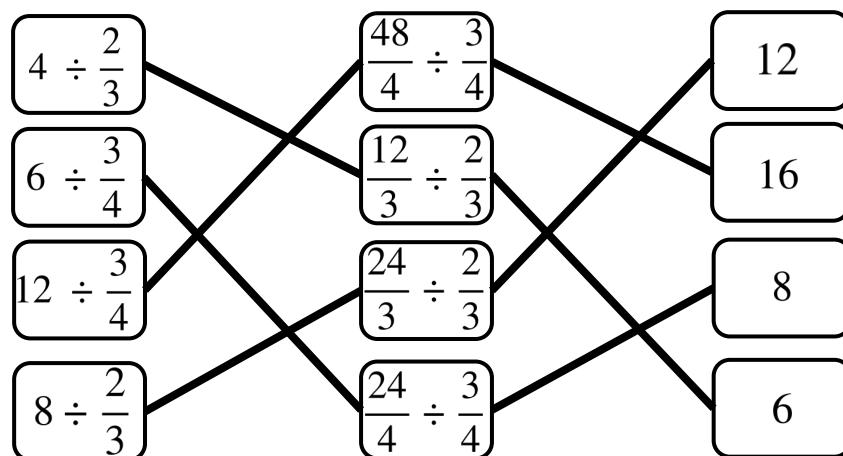
Task A: Making sense of dividing by a fraction

1) Use the bar models to show the division of the following.

a) $3 \div \frac{1}{3} = 9$ b) $2 \div \frac{2}{5} = 5$ c) $6 \div \frac{3}{4} = 8$



2) Match the question with the calculation and the answer.



3) Circle the following divisions which give an answer of four.

$3 \div \frac{3}{4}$

$5 \div \frac{4}{5}$

$4 \div \frac{2}{5}$

$4 \div \frac{4}{4}$

$12 \div \frac{2}{3}$

$12 \div \frac{3}{1}$

$8 \div \frac{1}{4}$

4) Put the following in ascending order, smallest to largest.

$3 \div \frac{1}{10}$ $\frac{30}{10} \div \frac{1}{10} = 30$	$10 \div \frac{2}{5}$ $\frac{50}{5} \div \frac{2}{5} = 25$	$3 \div \frac{1}{4}$ $\frac{12}{4} \div \frac{1}{4} = 12$	$10 \div \frac{2}{3}$ $\frac{30}{3} \div \frac{2}{3} = 15$
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$$3 \div \frac{1}{4}$$

$$10 \div \frac{2}{3}$$

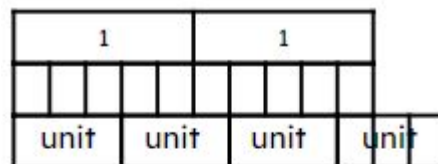
$$10 \div \frac{2}{5}$$

$$3 \div \frac{1}{10}$$

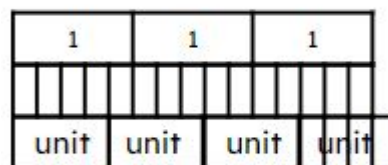
Task B: Deepening understanding of dividing by a fraction

1) Using the bar models, work out the answers to the following:

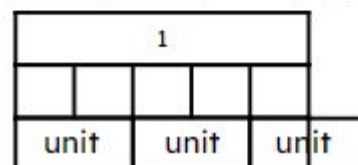
a) $2 \div \frac{3}{5} = 3\frac{1}{3}$



b) $3 \div \frac{4}{5} = 3\frac{3}{4}$



c) $1 \div \frac{2}{5} = 2\frac{1}{2}$



2) Izzy and Laura have done the following bar models for a question $3 \div \frac{2}{7}$

Both have made an error with their bar model.

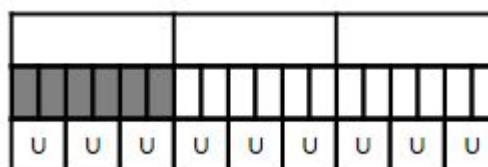
a) Explain where they have made their mistake.

Izzy has divided each 1 into sixths rather than sevenths.

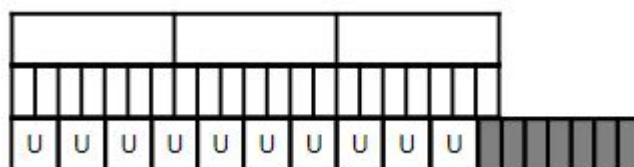
Izzy has calculated $3 \div \frac{2}{6} = 9$

Laura started to correctly use a unit to be $\frac{2}{7}$, but then incorrectly changed the unit to be $\frac{1}{7}$.

Izzy "The answer is 9"



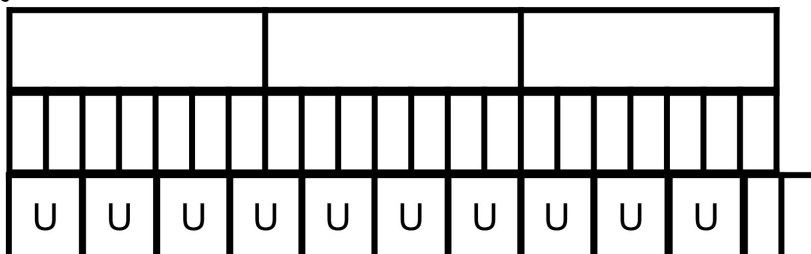
Laura "The answer is $10\frac{1}{7}$ "



b) Amend a bar model to help work out the correct answer.

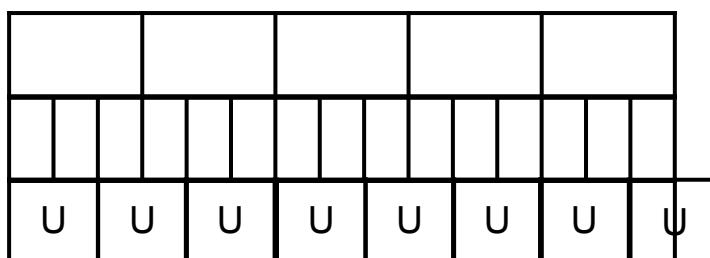
Changing Laura's so there is a consistency of units, the answer is seen to be

$$3 \div \frac{2}{7} = 10\frac{1}{2}$$

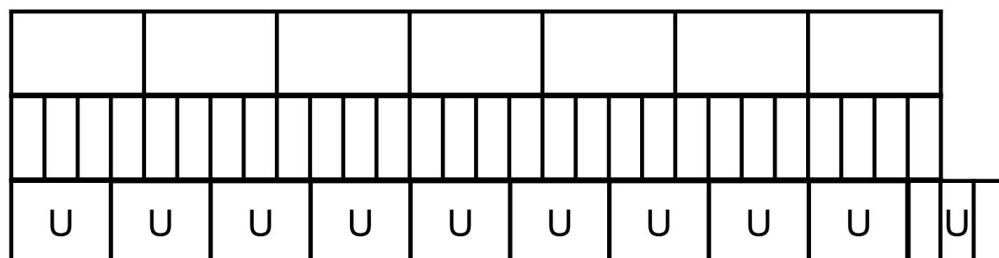


3) Using squared paper, draw your own bar models to work out the answers to

a) $5 \div \frac{2}{3} = 7\frac{1}{2}$



b) $7 \div \frac{3}{4} = 9\frac{1}{3}$



c) $5 \div \frac{3}{5} = 8\frac{1}{3}$

